STACKS, DISCRIMINANTS: LEMMA-FINITE-FLAT-TRACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 873-877 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 879-891, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 41 of queue_discriminant.json (pb=589, stmt 873-877, proof 879-891).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0BT5: sources/stacks-discriminant/pinned-original-source.tex, source lines 493–497. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a finite flat map of Noetherian rings. Then $\mathrm{Trace}_{B/A} \in \omega_{B/A}$ is the trace element.*

Proof. Suppose we have $A \to A_1$ with A_1 Noetherian and a product decomposition $B \otimes_A A_1 = C \times D$ with $A_1 \to C$ finite. Of course in this case $A_1 \to D$ is also finite. Set $B_1 = B \otimes_A A_1$. Since the construction of traces commutes with base change we see that $\mathrm{Trace}_{B/A}$ maps to Trace_{B_1/A_1}. Thus the proof is finished by noticing that $\mathrm{Trace}_{B_1/A_1} = (\mathrm{Trace}_{C/A_1}, \mathrm{Trace}_{D/A_1})$ under the isomorphism $\omega_{B_1/A_1} = \omega_{C/A_1} \times \omega_{D/A_1}$ of Lemma 0BT5. □

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.